

WORKED EXAMPLE

Continue backward from the output delta

Keep the original weights and the Lesson17 forward cache.

$$x = (1, 2), \quad t = 1, \quad \delta_1^{(3)} = -0.4533032667$$

$$y^{(1)} = (0.6681877722, 0.5744425168)$$

$$y^{(2)} = (0.5627638384, 0.4537407133)$$

$$W^{(2)} \text{ rows: } (0.4, -0.3), (-0.2, 0.2)$$

$$W^{(3)} \text{ rows: } (0.3), (-0.4)$$

The goal is four hidden deltas and all twelve hidden-layer parameter gradients. The three output gradients were calculated in Lesson18.

Each delta is the derivative of this one example loss with respect to one neuron preactivation. No batch averaging or parameter update happens here.

WORKED EXAMPLE

Sum downstream paths, then apply the local slope

Handout page 3, sections 5.3-6: every outgoing connection is a dependency.

$$\delta_j^{(l)} = \left(\sum_k w_{j,k}^{(l+1)} \delta_k^{(l+1)} \right) y_j^{(l)} (1 - y_j^{(l)})$$

The incoming backward derivative is also called the upstream gradient; it comes from downstream neurons in the forward graph. For each next-layer neuron k , multiply its delta by the connecting weight. Add all these path contributions, then multiply by the current neuron sigmoid slope.

$$\frac{\partial \ell}{\partial w_{i,j}^{(l)}} = y_i^{(l-1)} \delta_j^{(l)}, \quad \frac{\partial \ell}{\partial b_j^{(l)}} = \delta_j^{(l)}$$

A hidden neuron has no direct target label. Do not use hidden activation minus target as its delta. The y-t shortcut was specific to the output sigmoid plus BCE.

Use original connecting weights. Updating the output weights before computing hidden deltas would mix two different parameter states.

WORKED EXAMPLE

Hidden layer 2, neuron 1: delta

Multiply each downstream delta by its original connecting weight.

$$\text{upstream sum} = (0.3)(-0.4533032667)$$

$$\approx (-0.1359909800) = -0.1359909800$$

$$\text{local slope} = 0.5627638384(1 - 0.5627638384) \approx 0.2460607006$$

$$\delta_1^{(2)} \approx (-0.1359909800)(0.2460607006) = -0.0334620358$$

There is only one next-layer output neuron, so this upstream sum has just one term.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED EXAMPLE

Hidden layer 2, neuron 2: delta

Multiply each downstream delta by its original connecting weight.

$$\text{upstream sum} = (-0.4)(-0.4533032667)$$

$$\approx (0.1813213067) = 0.1813213067$$

$$\text{local slope} = 0.4537407133(1 - 0.4537407133) \approx 0.2478600784$$

$$\delta_2^{(2)} \approx (0.1813213067)(0.2478600784) = 0.0449423133$$

There is only one next-layer output neuron, so this upstream sum has just one term.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED EXAMPLE

Layer 2, neuron 1: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,1}^{(2)}} = y_i^{(1)} \delta_1^{(2)}$$

$$\frac{\partial \ell}{\partial w_{1,1}^{(2)}} \approx (0.6681877722)(-0.0334620358) = -0.0223589232$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(2)}} \approx (0.5744425168)(-0.0334620358) = -0.0192220161$$

$$\frac{\partial \ell}{\partial b_1^{(2)}} = 1 \delta_1^{(2)} \approx -0.0334620358$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 2 the sources are hidden-layer-1 activations, not the original inputs.

WORKED EXAMPLE

Layer 2, neuron 2: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,2}^{(2)}} = y_i^{(1)} \delta_2^{(2)}$$

$$\frac{\partial \ell}{\partial w_{1,2}^{(2)}} \approx (0.6681877722)(0.0449423133) = 0.0300299042$$

$$\frac{\partial \ell}{\partial w_{2,2}^{(2)}} \approx (0.5744425168)(0.0449423133) = 0.0258167756$$

$$\frac{\partial \ell}{\partial b_2^{(2)}} = 1 \delta_2^{(2)} \approx 0.0449423133$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 2 the sources are hidden-layer-1 activations, not the original inputs.

WORKED EXAMPLE

Hidden layer 1, neuron 1: delta

Multiply each downstream delta by its original connecting weight.

$$\begin{aligned}\text{upstream sum} &= (0.4)(-0.0334620358) + (-0.3)(0.0449423133) \\ &\approx (-0.0133848143) + (-0.0134826940) = -0.0268675083\end{aligned}$$

$$\text{local slope} = 0.6681877722(1 - 0.6681877722) \approx 0.2217128733$$

$$\delta_1^{(1)} \approx (-0.0268675083)(0.2217128733) = -0.0059568725$$

Both path contributions are included before applying the local slope.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED EXAMPLE

Hidden layer 1, neuron 2: delta

Multiply each downstream delta by its original connecting weight.

$$\begin{aligned}\text{upstream sum} &= (-0.2)(-0.0334620358) + (0.2)(0.0449423133) \\ &\approx (0.0066924072) + (0.0089884627) = 0.0156808698\end{aligned}$$

$$\text{local slope} = 0.5744425168(1 - 0.5744425168) \approx 0.2444583117$$

$$\delta_2^{(1)} \approx (0.0156808698)(0.2444583117) = 0.0038333190$$

Both path contributions are included before applying the local slope.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED EXAMPLE

Layer 1, neuron 1: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,1}^{(1)}} = y_i^{(0)} \delta_1^{(1)}$$

$$\frac{\partial \ell}{\partial w_{1,1}^{(1)}} \approx (1.0000000000)(-0.0059568725) = -0.0059568725$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(1)}} \approx (2.0000000000)(-0.0059568725) = -0.0119137449$$

$$\frac{\partial \ell}{\partial b_1^{(1)}} = 1 \delta_1^{(1)} \approx -0.0059568725$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 1 the source activations are the measured inputs $x_1 = 1$ and $x_2 = 2$.

WORKED EXAMPLE

Layer 1, neuron 2: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,2}^{(1)}} = y_i^{(0)} \delta_2^{(1)}$$

$$\frac{\partial \ell}{\partial w_{1,2}^{(1)}} \approx (1.0000000000)(0.0038333190) = 0.0038333190$$

$$\frac{\partial \ell}{\partial w_{2,2}^{(1)}} \approx (2.0000000000)(0.0038333190) = 0.0076666379$$

$$\frac{\partial \ell}{\partial b_2^{(1)}} = 1 \delta_2^{(1)} \approx 0.0038333190$$

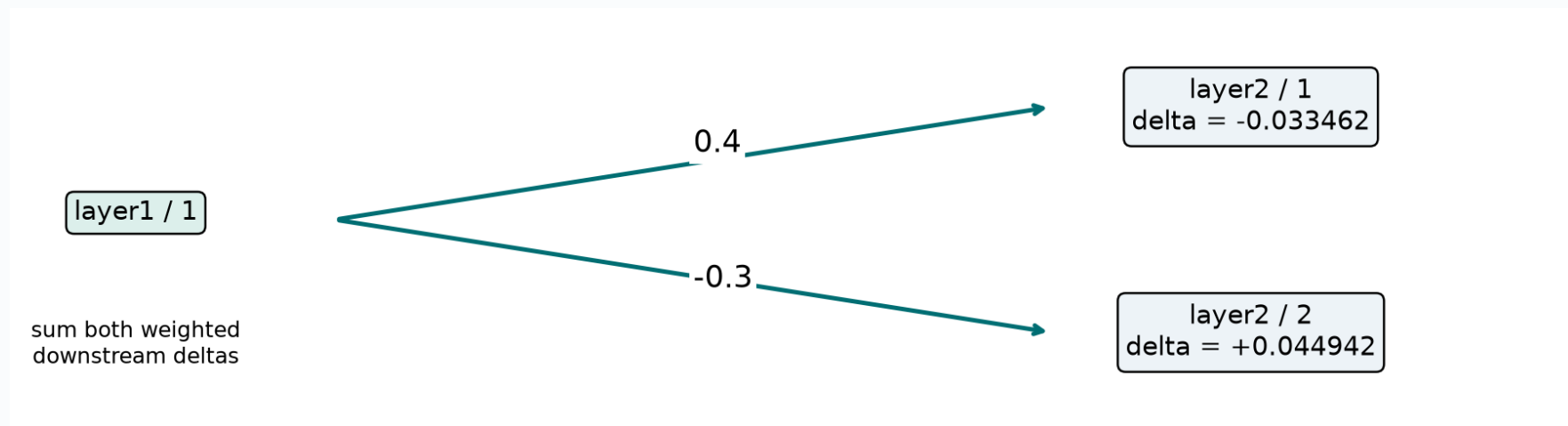
Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 1 the source activations are the measured inputs $x_1 = 1$ and $x_2 = 2$.

WORKED EXAMPLE

Why the first hidden neuron needs both paths

The diagram isolates its two outgoing connections to hidden layer 2.



$$0.4\delta_1^{(2)} + (-0.3)\delta_2^{(2)} = -0.0268675083$$

The two next-layer neurons both depend on this first-hidden activation. The chain rule adds the effects through both branches, even when a connecting weight is negative.

WORKED EXAMPLE

Checkpoint: every hidden parameter is covered

All values are derivatives of the same one-example loss at the same original state.

parameter	layer1 gradient	layer2 gradient
w1,1	-0.0059568725	-0.0223589232
w1,2	0.0038333190	0.0300299042
w2,1	-0.0119137449	-0.0192220161
w2,2	0.0076666379	0.0258167756
b1	-0.0059568725	-0.0334620358
b2	0.0038333190	0.0449423133

The twelve hidden gradients plus three output gradients cover all fifteen trainable parameters. Independent finite-difference checks of all fifteen agree.

INDEPENDENT PRACTICE

Your turn: propagate the target-flip delta

Use the original input (1,2), original parameters, but target $t = 0$.

$$\delta_1^{(3)} = +0.5466967333$$

The forward cache is unchanged, as shown in Lesson18. Recalculate both layer-2 deltas, both layer-1 deltas, and all twelve hidden-parameter gradients.

Write the separate downstream contributions at layer 1 before adding them. Then apply each neuron local sigmoid slope.

Explain why hidden deltas are not hidden y minus target, and why you cannot update an output weight before this calculation.

WORKED ANSWER

Practice: Hidden layer 2, neuron 1: delta

Multiply each downstream delta by its original connecting weight.

$$\text{upstream sum} = (0.3)(0.5466967333)$$

$$\approx (0.1640090200) = 0.1640090200$$

$$\text{local slope} = 0.5627638384(1 - 0.5627638384) \approx 0.2460607006$$

$$\delta_1^{(2)} \approx (0.1640090200)(0.2460607006) = 0.0403561744$$

There is only one next-layer output neuron, so this upstream sum has just one term.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED ANSWER

Practice: Hidden layer 2, neuron 2: delta

Multiply each downstream delta by its original connecting weight.

$$\text{upstream sum} = (-0.4)(0.5466967333)$$

$$\approx (-0.2186786933) = -0.2186786933$$

$$\text{local slope} = 0.4537407133(1 - 0.4537407133) \approx 0.2478600784$$

$$\delta_2^{(2)} \approx (-0.2186786933)(0.2478600784) = -0.0542017181$$

There is only one next-layer output neuron, so this upstream sum has just one term.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED ANSWER

Practice: Layer 2, neuron 1: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,1}^{(2)}} = y_i^{(1)} \delta_1^{(2)}$$

$$\frac{\partial \ell}{\partial w_{1,1}^{(2)}} \approx (0.6681877722)(0.0403561744) = 0.0269655022$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(2)}} \approx (0.5744425168)(0.0403561744) = 0.0231823024$$

$$\frac{\partial \ell}{\partial b_1^{(2)}} = 1 \delta_1^{(2)} \approx 0.0403561744$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 2 the sources are hidden-layer-1 activations, not the original inputs.

WORKED ANSWER

Practice: Layer 2, neuron 2: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,2}^{(2)}} = y_i^{(1)} \delta_2^{(2)}$$

$$\frac{\partial \ell}{\partial w_{1,2}^{(2)}} \approx (0.6681877722)(-0.0542017181) = -0.0362169252$$

$$\frac{\partial \ell}{\partial w_{2,2}^{(2)}} \approx (0.5744425168)(-0.0542017181) = -0.0311357713$$

$$\frac{\partial \ell}{\partial b_2^{(2)}} = 1 \delta_2^{(2)} \approx -0.0542017181$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 2 the sources are hidden-layer-1 activations, not the original inputs.

WORKED ANSWER

Practice: Hidden layer 1, neuron 1: delta

Multiply each downstream delta by its original connecting weight.

$$\begin{aligned}\text{upstream sum} &= (0.4)(0.0403561744) + (-0.3)(-0.0542017181) \\ &\approx (0.0161424697) + (0.0162605154) = 0.0324029852\end{aligned}$$

$$\text{local slope} = 0.6681877722(1 - 0.6681877722) \approx 0.2217128733$$

$$\delta_1^{(1)} \approx (0.0324029852)(0.2217128733) = 0.0071841589$$

Both path contributions are included before applying the local slope.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED ANSWER

Practice: Hidden layer 1, neuron 2: delta

Multiply each downstream delta by its original connecting weight.

$$\begin{aligned}\text{upstream sum} &= (-0.2)(0.0403561744) + (0.2)(-0.0542017181) \\ &\approx (-0.0080712349) + (-0.0108403436) = -0.0189115785\end{aligned}$$

$$\text{local slope} = 0.5744425168(1 - 0.5744425168) \approx 0.2444583117$$

$$\delta_2^{(1)} \approx (-0.0189115785)(0.2444583117) = -0.0046230925$$

Both path contributions are included before applying the local slope.

This delta belongs to the destination neuron. All incoming parameter gradients for that neuron reuse it.

WORKED ANSWER

Practice: Layer 1, neuron 1: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,1}^{(1)}} = y_i^{(0)} \delta_1^{(1)}$$

$$\frac{\partial \ell}{\partial w_{1,1}^{(1)}} \approx (1.0000000000)(0.0071841589) = 0.0071841589$$

$$\frac{\partial \ell}{\partial w_{2,1}^{(1)}} \approx (2.0000000000)(0.0071841589) = 0.0143683179$$

$$\frac{\partial \ell}{\partial b_1^{(1)}} = 1 \delta_1^{(1)} \approx 0.0071841589$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 1 the source activations are the measured inputs $x_1 = 1$ and $x_2 = 2$.

WORKED ANSWER

Practice: Layer 1, neuron 2: three gradients

Formula, substitution and value for both incoming weights and the bias.

$$\frac{\partial \ell}{\partial w_{i,2}^{(1)}} = y_i^{(0)} \delta_2^{(1)}$$

$$\frac{\partial \ell}{\partial w_{1,2}^{(1)}} \approx (1.0000000000)(-0.0046230925) = -0.0046230925$$

$$\frac{\partial \ell}{\partial w_{2,2}^{(1)}} \approx (2.0000000000)(-0.0046230925) = -0.0092461851$$

$$\frac{\partial \ell}{\partial b_2^{(1)}} = 1 \delta_2^{(1)} \approx -0.0046230925$$

Two ordinary incoming weights use two different source activations. The one bias uses constant source 1.

For layer 1 the source activations are the measured inputs $x_1 = 1$ and $x_2 = 2$.

WORKED ANSWER

Practice check: what changed and what did not?

The known target changed the output delta; all forward values stayed fixed.

$$\delta^{(2)} \approx (0.0403561744, -0.0542017181)$$

$$\delta^{(1)} \approx (0.0071841589, -0.0046230925)$$

All twelve hidden gradients were recalculated on the preceding pages. Their signs reverse relative to target 1 because the output delta changes sign and all other factors stay fixed.

Here every gradient is multiplied by the same ratio of new to old output delta. That is a check for this fixed-input, fixed-parameter, single-output example; it does not replace the chain-rule working.

Common mistakes: dropping a downstream path, using an incoming instead of outgoing weight in the delta, using a preactivation as a source, or changing weights before all gradients are ready.